LESSON PLAN

LOW SPEED FLIGHT (9)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	07
Total No. of Lesson Plans	13	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Pneumatic system basic design requirements, brief description, advantages and disadvantages of a typical pneumatic system.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Pneumatic system basic design requirements, brief description, advantages and disadvantages of a typical pneumatic system.
5. Specific Objectives	9.1.1.1 Trainee will be able to describe various Aircraft engineering systems.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of IB diagrams; Instructor supervision required for safety regarding high-pressure systems.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary source of power for aircraft pneumatic systems?
(b) Name one component used to store compressed air in the system.
(c) Why is compressed air preferred over hydraulic fluid for certain aircraft systems?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Basic design requirements for pneumatic systems • Brief description of typical pneumatic system components • Air cycle machine operation principles • Pressure regulation and distribution methods	L & Ds	PPT, IB	15 min
(b)	• Advantages of pneumatic systems in aviation • Disadvantages and limitations of pneumatic systems • Comparison with hydraulic and electrical systems • Safety considerations for pneumatic components	L & Ds	PPT, PDF	20 min
(c)	• Typical applications of pneumatic systems on aircraft • Maintenance requirements for pneumatic components • Troubleshooting common pneumatic system faults • Future trends in aircraft pneumatic design	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) List three key advantages of using a pneumatic system over a hydraulic system.
(b) Explain the primary function of an air cycle machine in the pneumatic system.
(c) Identify one major disadvantage of pneumatic systems related to energy storage.

12. Home Assignment and Follow Up	Prepare a brief note comparing the operational principles of pneumatic systems with those of hydraulic systems, referencing LP 08 content on control surfaces.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	A/c controls systems; mechanical, power assisted fully powered & fly-by-wire type of controls surfaces.	Refer to the master syllabus reference list	LP 08

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min